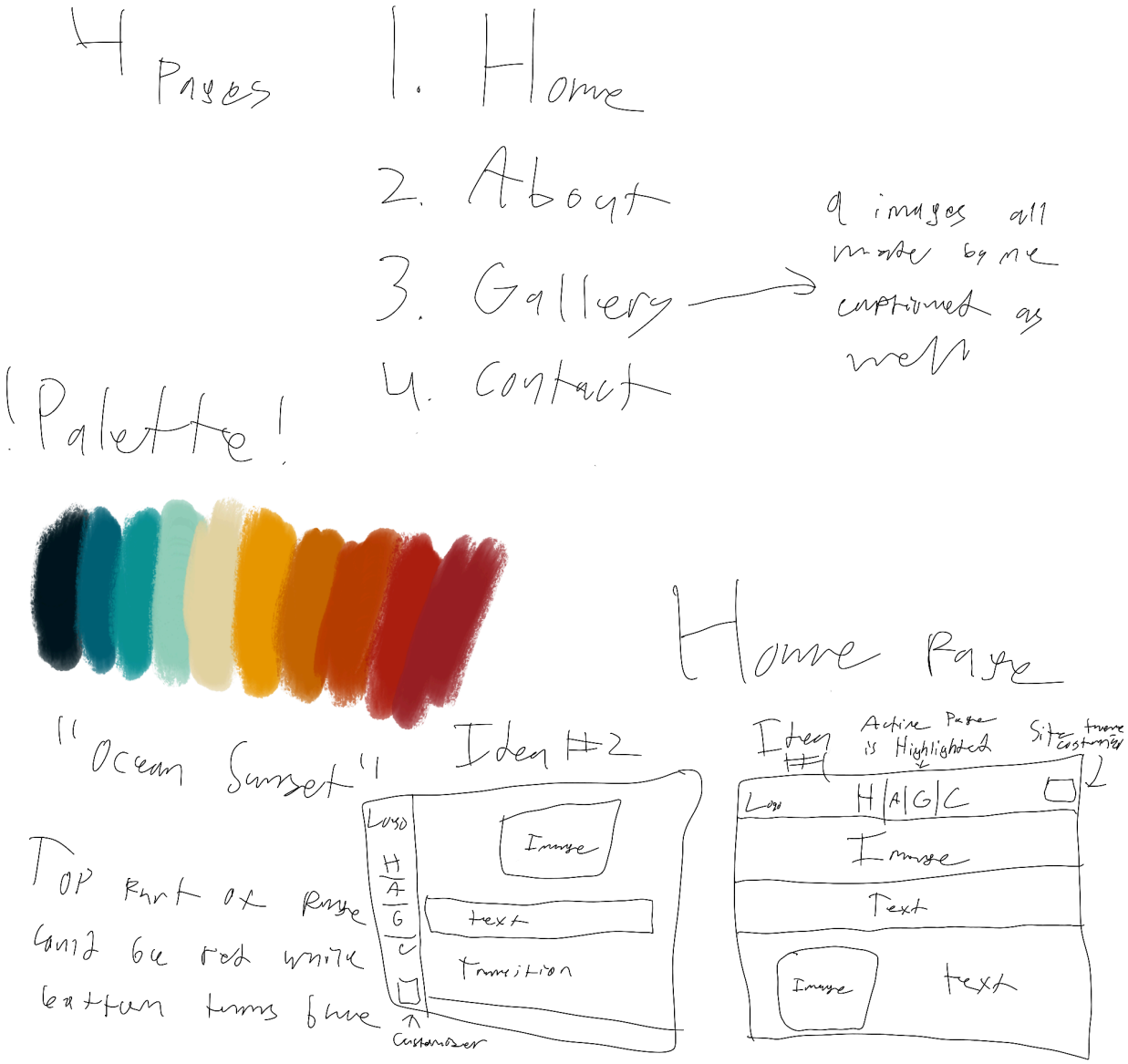
Predesign

For my website design, I knew that as someone with strong previous CSS experience that I had to make a website that I was personally proud of.

Initially, I wanted to start by designing the layout of the website.
I wanted to decide between a top header bar or a side bar as the main source of navigation. I ended up choosing the side bar because I believed it was a touch more unique, and could be better fitted to my “sinking” feeling of the website.



https://coolors.co/palette/001219-005f73-0a9396-94d2bd-e9d8a6-ee9b00-ca6702-bb3e03-ae2012-9b2226

I ended up choosing the “Ocean sunset” palette because not only do the more pastel colors compliment each other quite strongly, I realized I could make a very cool effect with my page. As the page continues down, the elements get darker and darker going from right to left on the palette, giving the “sinking” feeling and giving a very cool progression.

I ended up against adding a website theme customizer. Not because it would be particularly difficult, but because I wanted to implement my ocean sunset idea as the baseline, and replicating the “sinking” feeling would be difficult to do if someone chose a completely different theme that didn’t fit well with the design. I just have to make sure that the elements that aren’t supporting the color palette (like text and imagery) are higher contrast so they don’t end up blending into the page.

Starting the homepage

Before beginning, I had to make an important change. Joshua Comeau’s Reset.css is one of the strongest free files out there, it removes the default margins and spacing provided by browsers (and fixes webkit rendering as well). This makes the editing experience a lot easier.

I first started working on the left sidebar. I really want to work on its functionality for some reason but I shall wait. I first added the links I knew I was going to have to my 4 pages, Home, Gallery, About, Contact. I also added placeholder IMG tags with no source for 2 future buttons, the logo (which doubles for Home) and the Color changer which allows you to change it from “Default” (Ocean Sunset) to “Light” and “Dark” in case the specific colors I’m using are difficult to see.

First I set the top left image to be my online Profile Picture, and made it redirect to “index.html”
Then I set “Website theme changer”s margin-top: auto so that it would automatically maximize the margin between itself and the elements above it, forcing it down to the bottom where I wanted it.
Then I set my main clickable elements to margin: auto 0; which made it so that it automatically made itself equidistant from all elements above and below it. Now I have the basic layout of my left sidebar.

To make my sidebar look a bit nicer, i separated the distance of the 4 links by 50px each, and to make sure that I didn't add unnecessary margins I gave the first link (home) the class `.doNotSeperate` and added the CSS code:

```
header div > h1:not(.doNotSeparate)
{
   margin-top: 50px;
}
```

Which specifies that every h1 element that's directly under a div inside of the header element, that doesn't have the tag ".doNotSeperate" should have 50px of space added on top of it.

Problem 1: I realized for some reason despite my efforts to ensure that my left links remain centered, they were lifted up for some reason. I used inspect element and discovered that my Website Theme Changer margin auto was mixing with my links margin auto, removing the theme changers margin fixed this issue, and my elements are centered properly now. I also made my left sidebar a specific width instead of just 8% so that it would remain consistent across devices or browser resizing.

Problem 2: My links are very vertically thin, only going to their exact height as text. To fix this I set their minimum height to be the same as their width, 100%. But this made my earlier code to separate them funky and awkward, meaning I had to remove it. I realized that I wanted a little bit of space in between each link, so I repurposed the code.

```
header div > h1
{
   height: 100px;

   &:not(.doNotSeparate)
   {
      margin-top: 5px;
   }
}
```

Notice that instead of having 2 separate selectors, I could repurpose my tag so that I'm still not adding unnecessary margins while also setting the height of all of my divs. The & here denotes that when doing nested selectors, to append the child selector directly to the parent. Meaning instead of having h1 :not(), we have h1:not().

I also added "align-content: center;" so that the links text would be in the direct center of their elements.

To add some smoothness, I made it so when you hover on the left sidebar it increases its width to 200px (from 100px), this makes a nice looking sidebar (with a transition) that slides out as you hover over it. I want to make it so the rest of the website blurs when you hover over the left sidebar, but I will save that for when I have the rest of the website built.

Futureproofing
One of my plans is to add a color changer to the website in case the Ocean Sunset theme is too distracting. To do that, I would make it so that when the user clicks the theme changer icon that's in the bottom left of the screen, a little window pops up that allows them to edit their theme. The problem there is that if your mouse isn't hovering on the sidebar, it may inadvertently shrink the sidebar again, making for a clunky website experience. I'm predicting this might be an issue, and giving the header room for a class called ".headerActive" that has width set to 200px, so that in the future I might be able to implement some light JS that makes it so when the little theme changer menu is open, it gives the header that class preventing it from shrinking. To finish today, I wanted to add a google font that I liked. After some searching I found Encode Sans Expanded, a nice calm sans-serif font that gives the sidebar a more modern feature.



Website Screenshot #1

Continued Development

I want to flesh out my side bar before starting on my main content. If I can get that done I'll be slick.

To start, I wanted to try and make it so that the Home, Gallery, About, and Contact buttons extend out to say their full title when the bar is hovered on.

I wasn't initially sure how to approach this, since it sounds like a JS problem. But I realized that instead of changing the text directly, maybe I could just replace it with a span?

```
<h1 class="doNotSeparate">H</h1>
        <h1>G</h1>
        <h1>A</h1>
        <h1>C</h1>
```

Changed to

```
<span>
          <a class="doNotSeparate short">H</a>
          <a class="doNotSeparate long">Home</a>
        </span>
        <br>
        <span>
          <a class="short">G</a>
          <a class="long">Gallery</a>
        </span>
        <br>
        <span>
          <a class="short">A</a>
          <a class="long">About</a>
        </span>
        <br>
        <span>
          <a class="short">C</a>
          <a class="long">Contact</a>
        </span>
```

This way, upon hovering I can just hide the elements with "short" and show the elements with "long"
The problem I discovered is that by using display or opacity, I can have either a smoother transition or appropriate spacing, not both. Opacity causes the elements to continue existing after not being hovered, while display can allow them to be fully removed but doesn't offer any transition behaviors. My two other options were either custom animation or going really knee deep into absolute positioning, transition behaviors, and separate elements. Or maybe I didn't have to.

I also learned in my searches about the data-* tag. It allows you in html to store data of your specific choice. I could do data-short and data-long respectively equal H and Home so maybe I could use the attr function in css to change between them?

It turns out that using this along with the content: attr() function in css is also non-animatable, and also changes mostly the meta data associated with things, like i could change the href for example, but not the text itself. I'll have to give up on this idea for now, but I'm going to keep the instant transition from H -> Home instead of trying to make it smooth. Maybe it will grow on me.

Homepage: Main Content

First, I have to decide what I actually want on my homepage. Maybe just a basic introduction to who I am, my style of CSS, and as you scroll some small examples, and at the bottom a little blurb on "if you like my stuff contact me (as a link)"

I'm going to style a very bare bones layout of how I might want it first.

I started by adding a <main> tag offset by 80px, from the left to give room for the sidebar.

Inside main I added a few objects, a main container which will be the very first thing you lay eyes upon visiting the website, then below that a card container which will have 3 cards supporting my "business" or "portfolio."

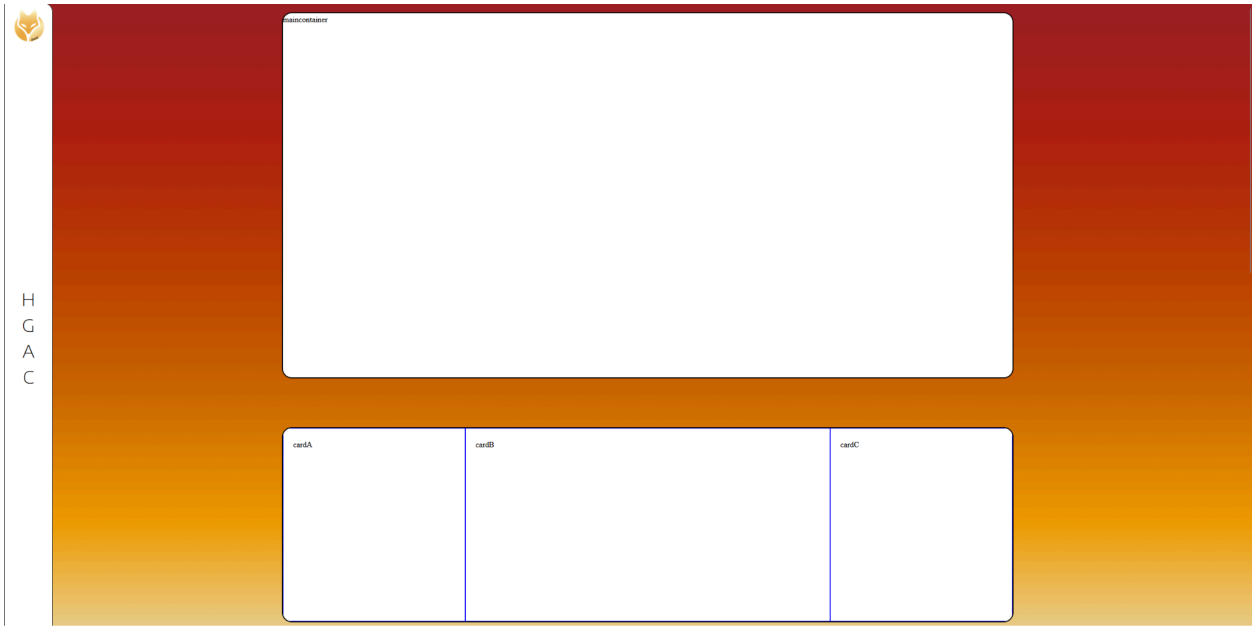
Past that, there will be an information container once again adding informational support for the content

Then finally, a contact container which will basically just say "ready to get started?" etc etc and direct the user to the contact page.

And then a footer which I may play around with or potentially delete.

Then I made a simple linear gradient incorporating all of the colors of my color palette from before, which stretch from the top of the page to the bottom. Right now it looks a little odd with the completely white containers but could look really beautiful if I can get the containers to blend well.

I also realized that I should probably separate my web pages so that styles.css doesn't have 6000 items to keep track of. I'll keep anything universal, like the header object, in styles.css while anything page specific like informationContainer will move to home.css.



Now, to add some more “effect” I wanted to blur the rest of the page when the left sidebar was hovered over.

I first added a fixed element that stays on top of the page no matter what, and when the header element is hovered, it activates the #blurElement object with the sister selector ~ to turn it from display: none to display: hidden, and I could then blur it using filter: blur;

A few problems with that

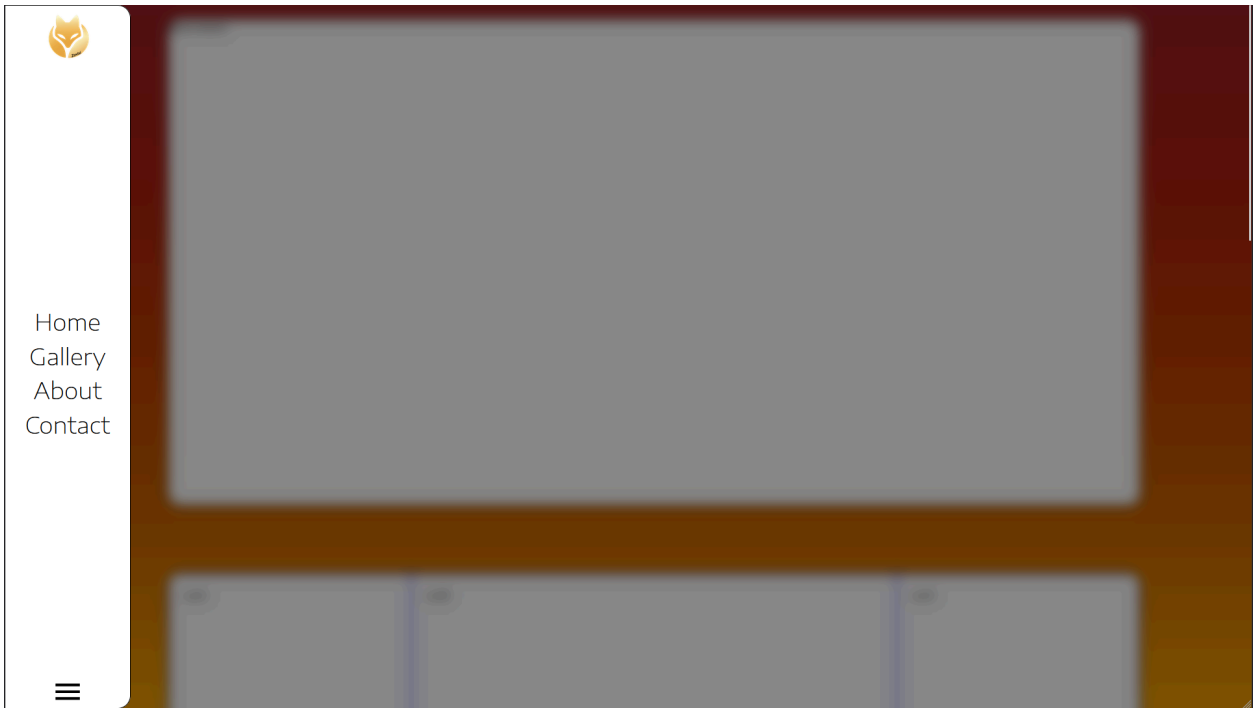
1. The element would snap instantly. I wanted it to transition from “normal vision” to “blurry and slightly darkened” to highlight the left side, and not to snap as to not alarm the viewer
2. The element would prevent the user from clicking on the main page if display: hidden; was not used. Meaning I had to somehow prevent the element from existing when the user wasn’t hovering on the element, but couldn’t use display or visibility because those don’t have transition properties.

First, I tried with opacity. It worked fine initially but the main issue is that the opacity didn’t help me with the blur effect. I actually settled for setting the background as a completely clear RGBA of (0, 0, 0, 0) while inactive, and turn it into (0, 0, 0, 0.5) when active, and adding a transition state to that meant upon hovering on the left header would turn the main page slightly dark.

To fix the issue of the element blocking the user from interacting with the webpage at all, I learned that pointer-events: none; could be set, preventing the user from interacting with that website at all in the first place. (from w3schools)

And finally, instead of using blur on the element, I actually had to use backdrop-blur (and the webkit variant too) because blur only blurs the element itself, not anything behind it, but backdrop blur blurs anything behind the element.

After diagnosing those problems, I'm met with some of the smoothest designs I've made so far. It looks very beautiful when hovering, and I'm in love with the blur effect. Great choice if I do say so myself.



I learned about scroll driven animations from
https://www.joshwcomeau.com/animation/scroll-driven-animations/
Josh w Comeau who is someone i regard highly about css, I've applied scroll driven animations in a similar fashion to my linear gradient background to my left header, so that as you scroll down the left sidebar changes color. I like it, but I think the linear gradient background clashes with the general feel of my website. I think I'm going to attempt to change it to a custom color to each level to make it more modern feeling.

Apparently you can use instant-stop double-definitions on the linear gradient to try it out.

I thought of something genius here. What if I make the left sidebar see through but blurred, like apples liquid glass effect?

I removed the custom background animation completely, and I gave the backdrop filter blur to the header. The problem right now is that the element that blurs and darkens the rest of the page is affecting that element too, making it too dark. I figured that I can just make it brighter using the brightness filter at 200% which turns out to work pretty well. Looks great!

I have a new problem though. I'm not in love with this background. It sort of looks like a setting sun, yes but it's rather loud.

I'm thinking I want to pick a nicer, simpler looking color scheme. First, I'm going to set everything back to the color white.

I kind of liked the **Soft Lavender** color, but I had an idea. I wanted to include neumorphism into my design and I think I know how to do that.
https://coolors.co/palette/22223b-4a4e69-9a8c98-c9ada7-f2e9e4

First I re-added my custom linear gradient code and added the double stops again while only using the few Soft Lavender colors, it looks very nice and has lots of potential I think

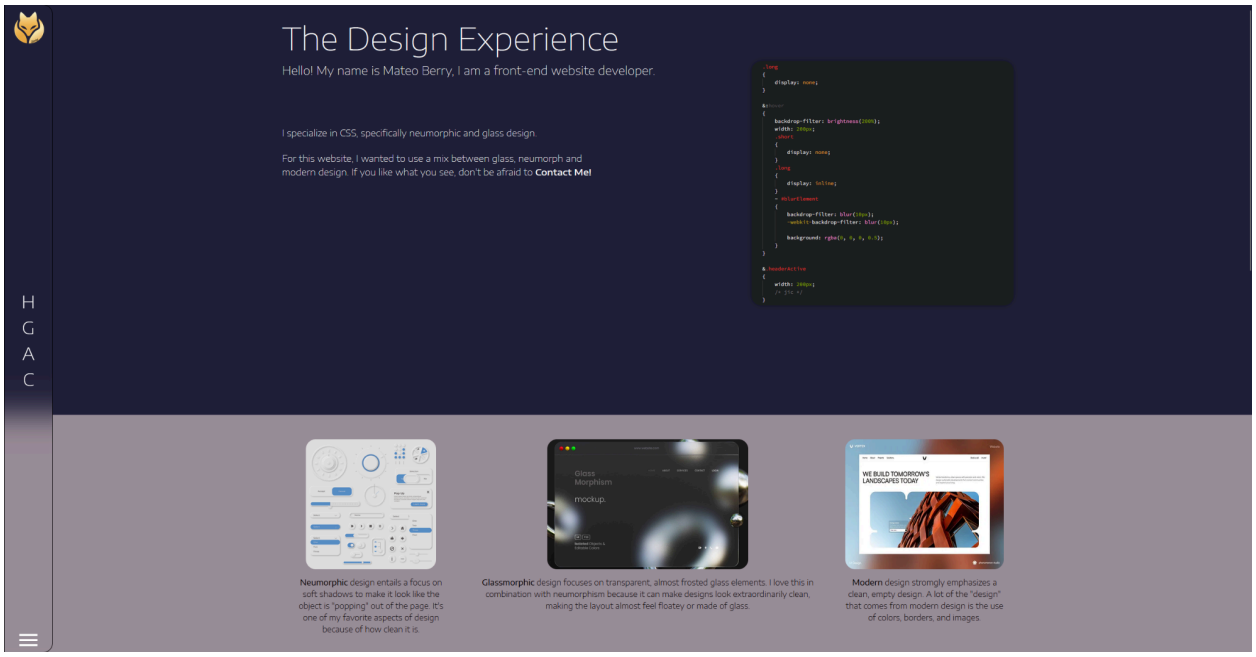
I've done neumorphism before, and the key is good box shadows.

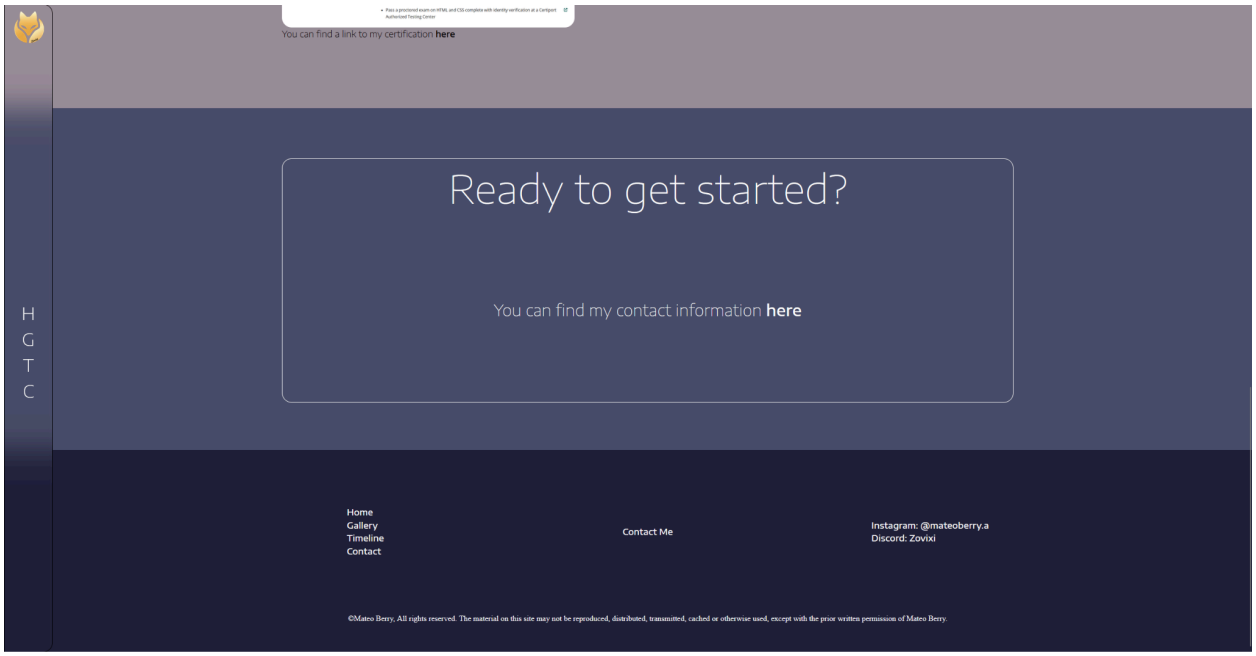
After messing with box shadows a bit, I figured the best box shadow I could ask for is a very thin, nearly transparent one with pure black, as well as an inset of the same thing. It gives *a* neumorphic design but not really the design I was purely thinking of, although I cannot find the design I'm thinking of anywhere so maybe it's not possible using standard css.



So far, I'm really liking how I've designed this so far. It has a lot of potential.

To continue my webpage, I wanted to make the design a little more responsive. I removed the neumorphic raise from my elements and instead made it so that upon hovering on an element, it applies that raised look. I do like the look. I also threw in some images and text to better fit my portfolio





Finally, I continued adding content to the rest of my boxes. Information which contains my credentials, contact which is honestly just to make use of more space, and the footer which contains navigation, the contact me button again, and some of my socials.

The first page is finished! The rest of my pages will mostly be the same, but with their own unique structure.

I plan for Gallery to just have a square of 9x9 images, if you hover over an image it will enlarge automatically like a pull out card, and show you a little caption describing the image with it.

For the timeline, I want to make a small timeline of my education (which is still growing !), i can easily make this in a photo editing software and import it to my website.

And for contact me, it will of course be a custom form.

The backgrounds of each page could be one of the colors I used on the main page as well, which will make dealing with any styling a lot easier.

First to work on the gallery, I literally just duplicated the index page. A little lazy but honestly saves me a lot of time from pasting everything over and reconfiguring everything.

After moving everything over, I decided to pick a light background for my gallery.



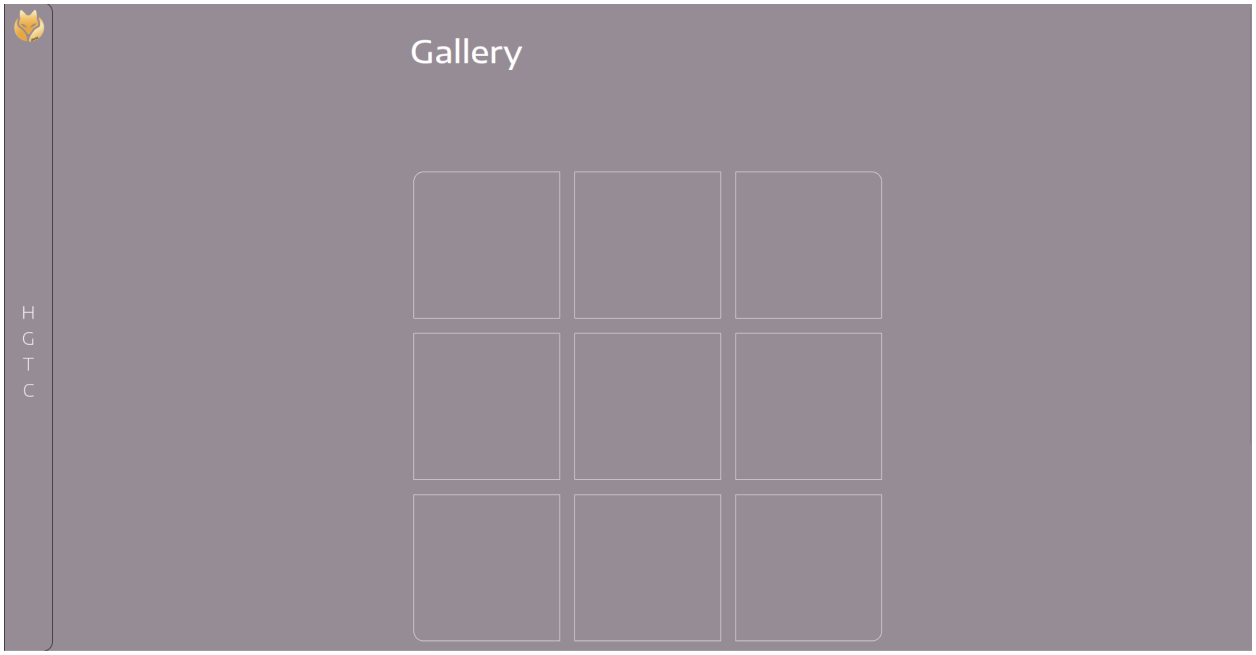
Perfect. Now to add a main section.

I knew that I had to use Grid for my items. This page honestly should be pretty easy to make.

While working on my grid of items, I realized I wanted to make it look like a centralized container but a traditional border radius wouldn't do.
I needed to implement a class based system for corner border radiuses

```css
.topRightBorder
{
    border-top-right-radius: 20px;
}
.topLeftBorder
{
    border-top-left-radius: 20px;
}
.bottomRightBorder
{
    border-bottom-right-radius: 20px;
}
.bottomLeftBorder
{
    border-bottom-left-radius: 20px;
}
```

Sweet!



Works well!

I wanted to make each image on my website clickable and it would make it center stage and have a caption present there however… that requires javascript and unlike my exercise i don’t want to go about ripping code from other people for my lab.



After messing with a stock photo, I found that figure + figcaption could work well as a replacement. It takes up the box it’s in and sits directly below the image and I don't have to work a lot with trying to somehow put it in between the grid gaps.

I figured out though that as long as i define the image to specifically take up 100% of the height and width of the container, it would allow me to kind of “push” my captions below the image which ended up looking fine! Although the images started clipping out of bounds of their parents so i learned about object-fit: cover; which essentially told the image to clip anything outside of the scale of it’s bounding box.



I finished with all of my images and captions! No issues here.

Time to move on to the timeline.

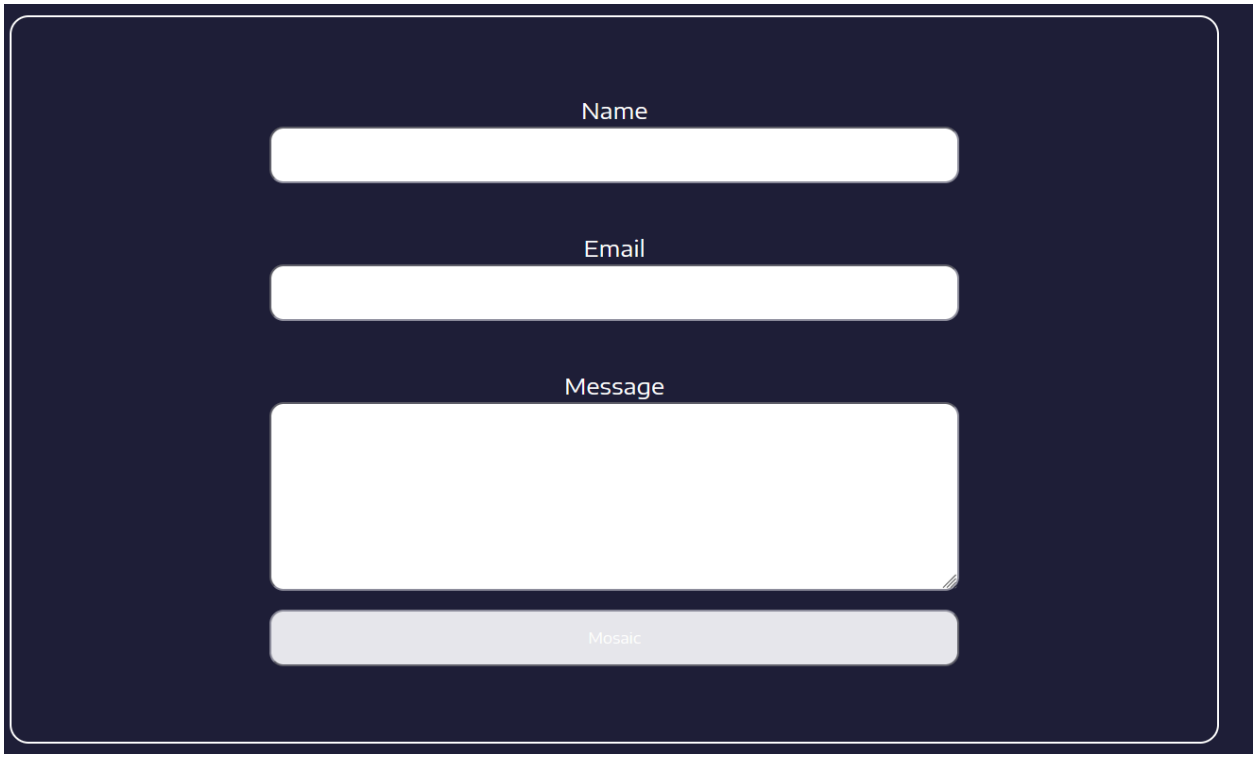
I figured I could make the timeline look good without working on it with CSS for 5 hours, so I went ahead and made an image of a timeline in a photo editing program so that

[insert timeline stuff here when done]

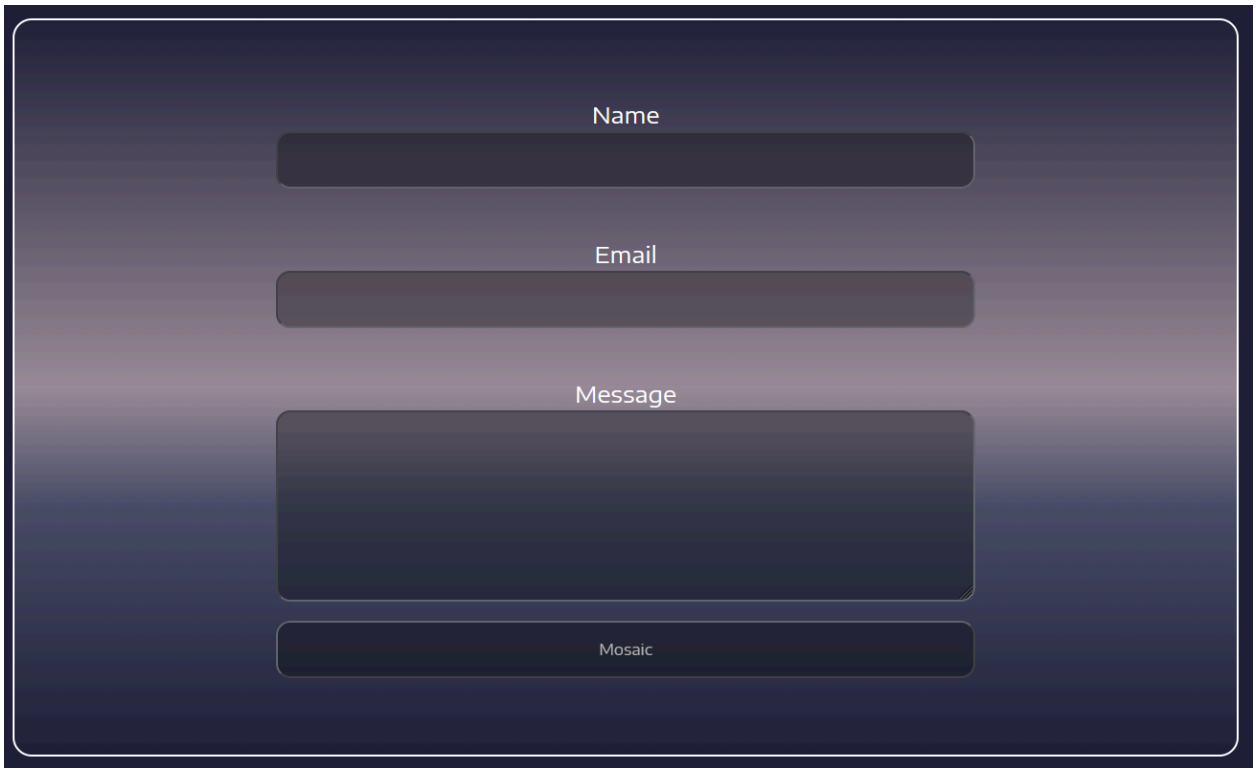
Now that I'm done with that, I can move on to my last item. A contact form!

This shouldn't be too hard… I think,,

First I started with my basic page layout from before, then I added a general form box.



Now to add some custom design



And wow holy smokes I cooked with this a little too hard. This is extraordinarily beautiful, I was just messing around with some basic glassmorphism and somehow invented this thing.

Now that the main content of the website is finished, I wanted to add a mobile version of the website. The problem is, I don't have any way of moving these items to a mobile device quickly or easily, and my browsers device emulation developer tool works for resizing the viewing window but no changes take place because it's still reading my device as a computer, even when I request a mobile website. I cannot accomplish this task because of my personal restrictions, and I apologize for that.

Part of my documentation I'm required is for accessibility. Throughout my journey I've added alt text to all my images since that was drilled into me as a w3c standard, so that is an example of my accessibility features, but another notable feature i noticed was that non-mouse users couldn't access the left buttons, so if using TAB you can tab over to the implement anchor tags to switch websites without having to hover over the items to activate the links!

Reflection

Building this website was a long but interesting journey. Although I was unable to style the website for mobile users properly (due to a lack of tools) and I didn't have the javascript know-how to implement custom website themes, this project allowed me to explore a lot more into the specifics of css design. I learned a lot about the differences for flexbox and grid, especially aligning items which is always a pain for me. I discovered neat effects like backdrop-filter and blur. Another big one was the discovery of more specific css functions, like :not or :has, those are vital if you want to make great looking response websites. The HTML portion was big too. I figured that semantics make not only navigating your page through the code easier, but managing it across several css files easier too. My proudest designs are the left sidebar, which I absolutely love the glass blur hover effect it applies to the entire page when you hover over it, and the essential design of the contact form, because that applies glassmorphism as well but makes use of gradient colors and all around just looks super neat. The late implementation of the seed constraint was definitely… constraining to say the least. But I did manage to make it through.
Overall, this was a fun and useful project to help improve my native website building skills, and hopefully it will allow me to expand my knowledge and my site accessibility through javascript when we learn that in the future, although the constraint with time, the seed, and the general educationally-related information curve is definitely not for newer or less comfortable students!